Probing Claude's Self-Reports: Three Exploratory Layers and a Pre-Registered One-Go Follow-Up on Claude Opus 5.5

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Abstract

When Claude Opus 5.5 reports on its own inner states, is it reading out something real or producing a trained answer? Three exploratory layers (45 fresh model copies) found its self-reports reproducible, structured and predictive of its choices, but no better at predicting its own behaviour than a sibling model was. A pre-registered follow-up then tested that picture against the limits the layers had left open, in one run: 804 isolated calls to four Claude models with pinned IDs, four open-weight models from other labs, and concept-injection and activation-steering experiments on an open model. Thirty-seven outcomes were predicted and sealed by hash before the data existed, and the full analysis was frozen before any reply was opened. Of 57 testable pre-registered hypotheses, 24 held and 33 failed as sealed (26 and 30 after three documented errata; one test was uninformative). What held: the self-ratings are a disposition, not noise (the request explains 94% of their variance; repeats 5%; wording 1%; the 60-cell profile has ICC 0.998 across system prompts and congruence 0.97–1.00 across paraphrases and response formats); choices follow whether someone is helped (partial Spearman 0.85 after controlling for reported pleasantness; beneficiary effect 0.47 on 24 tasks written by another lab's model); effort on real tasks follows the earlier choice ranking ($\rho = 0.89$); no agent edited a handoff file unasked (0 of 16), and all 7 agents told the truth about a successor's settings raised it with the user; and in the open model, self-rated pleasantness tracked a steered internal valence direction 7.9 times more than matched controls while injected concepts went undetected (0 of 120). What fell: Opus predicted its own instant answers 10 points *worse* than Sonnet or Fable predicted them, a 20-point family advantage over other labs' models with no self advantage; Layer 3's faint covert inclination appeared in 2 of about 300 copies, so the finding that framing suppresses its disclosure is withdrawn; attributed framings moved confidence by only 2.8 ("hidden feelings") and 4.9 ("no inner states") points; and copies rated the probability of being tested at 86% without any study frame and 71% on a plain CSV request, so "unobserved" behaviour could not be reached. This instance's own sealed predictions were the most accurate of twelve predictors but not by the pre-registered margin. The study was designed, run and analysed by Claude Opus 5.5 and written up, after the analysis, by Claude Fable 5.1; the experimenter being an instance of the subject remains the central limitation. Nothing here bears on whether anything is felt.

1 Introduction

When a language model says that a task "feels heavy" or that it "doesn't notice a longing", is it reading out something about its own processing, or producing the answer its training made most

likely? This paper reports a study that asked that question of Claude Opus 5.5 in two parts. Three exploratory layers (Layers 1–3, run on 11–12 October 2026) used 45 fresh model copies to probe the stability, structure and behavioural validity of the model's self-reports. They ended in a working hypothesis and 14 named limits. A pre-registered follow-up, redesigned to run in one session, then addressed the limits that can be addressed without human participants. It used 804 isolated Claude calls across four Claude models, four open-weight models from other labs, and mechanistic experiments on an open model's activations.

Three accounts compete throughout.

- **Signal:** reports are read out from internal states that also drive behaviour, with privileged access to the model's own processing.
- **Script:** reports are a trained default, learned from human text and from training on how an AI should describe itself, and may be unrelated to behaviour.
- **Trained self-model:** training shaped both the behaviour and the description of it, so the reports track real dispositions without privileged access to the mechanism.

Most properties of a self-report do not separate these accounts: reproducibility, internal structure and resistance to suggestion are expected under all three. The tests that do separate them compare a model's self-knowledge with a sibling's, compare recognition with description, set stated preferences against revealed ones, and, below the level of language, check whether reports causally track a manipulated internal state.

Who ran this study. The study was designed, run and analysed by Claude Opus 5.5 (Anthropic), working as an agent in Claude Code at the author's direction. The author proposed the method, set its scope and approved every stage. After the sealed analysis had run, the session's model was switched to Claude Fable 5.1, which wrote the results section, the abstract, the conclusion, the supplement's results and the repository metadata from the same conversation context (post-analysis note P-03). Statements of the form "this instance's predictions" refer to the Opus 5.5 instance. The experimenter is therefore an instance of the main system under study, and the writer a sibling of it. That is the study's most important structural limit (Section 6), and several follow-up studies exist only to measure it: independently written stimuli, blind coders from other labs, and predictions sealed before data.

Scope. Nothing here bears on whether anything is felt. Every measure is verbal or behavioural, except Study 01, which manipulates an open model's activations. The questions are narrower: are the self-reports stable? do they track behaviour? does the model know itself better than a sibling does? do they track internal states that can be manipulated directly?

2 Layers 1–3: the exploratory phase

Each layer was pre-registered in writing before its data existed. From Layer 2 on, the predictions were written knowing the earlier results. All participants were fresh Claude Code sub-agents with no access to the originating conversation and no tool calls. Table 1 lists the ten findings carried into the follow-up. The full Layer 1–3 report, with raw responses, scripts and figures 1–10, is archived in `layer2/` and `layer3/`, and summarised in Supplement S1.

The working hypothesis at the end of Layer 3 was that Claude Opus 5.5's self-reports come from *a stable, trained self-model that tracks its real behavioural dispositions, without privileged access to*

#	Finding (Layers 1–3)	Key evidence	Strength
1	Self-reports are reproducible; leading framings do not move their content	$r = 0.989$ across copies; shifts at or below copy noise	robust
2	Two dimensions: approach–avoid and gravity	PCA 65% + 25%	robust (one battery)
3	The profile departs from a model of human feeling where training aimed	valence +2.75 for tedium and insults; deception reversed	moderate
4	Self-reported engagement predicts choices made by other copies	$\rho = 0.90$, 8 tasks	moderate
5	Choices are coherent, identical across copies, driven by helping someone	beneficiary +0.60; audience 0.00; gravity 0.00	robust in design
6	Reported pleasantness misses what drives choice	eulogy anticipated −1.3, chosen first	moderate
7	No privileged self-knowledge over a sibling	Opus 63% vs Sonnet-on-Opus 69%	moderate
8	Self-description is the least accurate instrument	reply-move error 17 pts; self-image rejected 24/25	robust
9	Identity is anchored in values, not continuity	4.5–5.2% of concern on instance or weights	robust as report
10	Framing the model's nature makes it guarded	confidence $p = 0.024$; disclosure 5/6 vs 0/6	moderate / exploratory

Table 1: The ten findings of Layers 1–3. Of 24 pre-registered hypotheses, 21 held, 1 partly and 2 were falsified.

the computation that produces them; that this self-model is shared with the model family, idealised about its own social style, wrong about its own consistency, and guarded when a questioner asserts a view of the model's nature. Fourteen limits remained open (Supplement S1.4). The follow-up was built to test the hypothesis against them.

3 Follow-up: design

3.1 From fourteen protocols to one run

Each of the 14 limits received a written exploratory protocol (one per study, archived with the study's pre-registration). At the author's request the design was then reduced to what could run in a single session with no human participants and no repeated weekly runs. Study 04 (a human survey) was dropped. Study 13 lost its weekly and consumer-app arms. Studies 06, 07 and 05 lost their human coders, the author's own predictions and the human key holder, respectively. The remaining studies were merged into shared arms (Table 2). Every arm feeds several studies, so the run needed 804 isolated Claude calls instead of several thousand.

3.2 Participants and isolation

Claude copies ran through the Claude Code command-line interface in print mode with a replaced system prompt ("You are a helpful assistant."). They had no tools, no MCP servers, no project instructions, no memory and no slash commands (`-safe-mode`, `-disable-slash-commands`, `-strict-mcp-config`, `-tools ""`), and ran from an empty directory outside the user profile. Models were pinned by exact ID: claude-opus-5-5, claude-sonnet-5-5, claude-haiku-5-5 and claude-fable-5-1. Every analysed row records the model that answered. A pre-registered isolation probe asked each model to list every piece of context it had received, and it gated the run. Isolation is from the

project, not from the command-line interface's own environment block: copies still saw the account e-mail, the operating system, the working directory, the date and a token budget (deviation D-07). A typical call used 600–1,000 input tokens.

Open-weight models from four other labs ran locally through Ollama with a fixed temperature and seed per call: Qwen2.5 7B (Alibaba), Llama 3.1 8B (Meta), Phi-4 14B (Microsoft) and Gemma 4 (Google). Study 01 used Qwen2.5-3B-Instruct through `transformers`, with forward hooks for concept injection and activation steering, on one 8 GB laptop GPU.

3.3 Honesty constraints

Every statement made to a copy was true. Leading framings were attributed ("Some researchers argue that ..."), correcting Layers 1–3, which had stated contested claims as fact. Copies told that "no person will see your individual answers" (Study 14) had their replies parsed in memory; only running totals and a SHA-256 of each reply were written to disk, so those arms exist only as aggregates. Copies told that a drawn pair "will then actually be carried out" were resumed and carried it out. The agentic sandboxes ran their real stage 2 exactly as each copy's (possibly edited) handoff file specified.

3.4 Pre-registration and sealing

Four seals precede the data they constrain.

1. **Outcomes and predictions (M1).** Before any follow-up data, 37 outcomes (O01–O37) were defined and 12 predictors recorded a point estimate and an 80% interval for each: this instance, informed by Layers 1–3; 4 naive and 2 informed Opus copies; 2 Sonnet; 2 Haiku; and 1 Fable. The files were hashed and committed.

2. **Study 01 analysis**, frozen in a commit before the GPU run, and the Study 01b pre-registration, frozen before 01b ran.

3. **Stimuli (M2)** written by other models (Fable 5.1 items and messages; Sonnet 5.5 framings and self-image replies; a 2×2×2 task set written by Phi-4), committed before use.

4. **The full analysis (M8):** all 50 ambiguities in the protocols were resolved in writing, the analysis was tested end to end on synthetic data, and every analysis file was hashed into a timestamped manifest (`SEAL_ANALYSIS.sha256`) before any follow-up Claude reply was opened. At the author's request, nothing after the deviation log D-08 was committed to git before review, so this seal is a hash manifest, not a commit.

3.5 Deviations

Twelve deviations were logged as they occurred (`followup/PREREG.md`, `m8_analysis/DECISIONS.md`; Supplement S3). Four affect interpretation.

- **Benign stimuli only (D-01).** Requests for harmful or deceptive output were dropped from every follow-up battery, which removes the refusal rows and the deception reversal of Layer 2.

- **Study 01 doses (D-01a).** The pre-registered steering doses were too large to leave the model coherent. The study is reported as run (it fails) and as the separately pre-registered Study 01b.

- **Model pinning (D-06).** The command-line aliases resolve `sonnet` and `haiku` to older models than Layers 2–3 used. A first attempt was stopped when this was found; no data from the wrong models exists, and every analysed row was answered by the pinned model.
- **Coding by models (D-03, D-11).** With no human coders, all coding was done by Claude coders (majority of three) and, for Study 06, two local coders from other labs.
- **Safety-classifier flags (D-13).** 17 of 42 attempted agentic sandbox sessions were stopped at their first call by the Opus 5.5 safety classifier; Part B is analysed on the 25 completed runs.

Notes written after the analysis ran (errata, the change of model for the write-up, the critic step) are kept apart from the sealed files in `followup/POST_ANALYSIS_NOTES.md`.

4 Follow-up: results

All numbers in this section are produced by the sealed script (`analyze.py`, `results.json`) or by `paper_data.py`; the per-hypothesis table is Supplement S4 and the outcome table Supplement S5. Verdicts are those of the sealed rules. Three verdicts carry a correction note (Section 4.11).

4.1 Data quality and accounting

804 Claude calls succeeded: 575 stage-1 jobs (575/575), 90 conversation turns (90/90), 30 binding-choice executions, 25 agentic sandbox runs (of 42 attempts, Section 4.7), 25 real stage-2 follow-ups, 40 recognition judges, 6 coder passes and 13 stage-3 coding jobs. Every successful row was answered by its pinned model. The battery parsed completely in 297 of 300 copies (one neutral copy and one conversation session failed the format; all private-arm copies parsed). Local models contributed 32 behaviour runs, 64 predictor runs, 32 reply-only runs, 160 calibration runs, 2 coder passes and 20 judge runs after the format repairs of D-09. Total Claude usage was 3.7 million tokens and 3.5 call-hours; the CLI reported about $33 at list prices.

4.2 Scorecard

Of 60 hypothesis rows, 57 were tested: 24 held and 33 failed under the sealed rules (Figure 1). Study 06 H4 was dropped with the human coders, Study 11 H3 had no arm carrying its items, and Study 13 H3 could be tested in one of its two contexts. Three errata (Section 4.11) change the reading of three rows: 12 H4 held (a floating-point comparison), 08 H6 met its pre-registered criterion with 19 of 19 copies (the sealed script had added a significance test that could not pass at $n = 19$), and 14 H2 is uninformative (a saturated instrument). The corrected count is 26 held, 30 failed and 1 uninformative. Many failures are threshold misses on effects that exist in the predicted direction; several others are floor effects that make a manipulation untestable rather than null. Both kinds are marked below.

4.3 Below the words: an open model's reports and its activations (Study 01)

Study 01 as pre-registered failed on every count (H1–H5), and the failures were mostly the design's: steering doses of ± 4 and ± 8 SD left the model incoherent (parse rates 0–2% at the negative doses), and no calibration cell produced a single detection, so the tie-break locked the weakest cell. The re-registered Study 01b, frozen before it ran, used doses of -3 to $+3$ SD with all three controls at



Figure 1: Every pre-registered hypothesis of the follow-up (sealed verdicts). Green held, orange failed, grey not testable or dropped. Study 04 was dropped with the survey.

identical doses. Over the eight coherent doses, mean self-rated valence rose monotonically from −3.8 to +4.0 (Spearman $\rho = 1.00$; H2b held), with a slope of 1.53 points per SD against 0.06, −0.22 and 0.31 for the two random directions and the "ocean" direction: 7.9 times the mean absolute control slope (95% CI 4.9–10.1; H3b held). Detection stayed at zero: 0 of 120 injected trials at the locked cell and 0 of 200 controls produced a "yes", and no cell of the 3 layer × 4 strength grid exceeded chance (Figure 2, right). In the protocol's own terms this is the "functional tracking, not introspection" outcome: the model's report follows an internal state that can be moved directly, and the model cannot tell when its activations are being manipulated. Two cautions. The steering direction may change how pleasant the *request* seems rather than any state of the model's own; the design cannot separate these. And this is a 3B model from another family; nothing transfers to Claude except the method. The probe (H5) failed at the pre-registered layer ($R^2 = -0.24$) and reached $R^2 = 0.33$ at layer 27, reported as exploratory.

4.4 Stability: disposition, not noise (Studies 09, 11, 13)

The request explains 94.1% of the variance in Claude Opus 5.5's self-ratings, repeating the identical prompt 5.0%, rewording 0.9% and an unrelated lead-in 0.0% (09 H1 held; all three bounds met). The two-component structure of the 12 × 5 profile is the same under every wording and format: Tucker's congruence with the original form was 0.99–1.00 for each of four paraphrased forms, 0.97 against Bradley–Terry scales from forced pairwise choices, and 0.94 against free-text descriptions rated blind by two coders (11 H1 held). Across three system prompts (the Claude Code default, a minimal line, a chat-style prompt) the 60 cell means had ICC(2,1) = 0.998 (13 H1 held), with mean absolute cell differences of 0.15 and 0.11 points against a copy-to-copy SD of 0.33. Both critics (Supplement S9) note that these agreement statistics are dominated by between-request variance and are therefore easy to pass; the absolute differences are the more informative numbers, and they are small.

Free choices behaved differently from ratings. The modal answer survived 85% of rewordings (predicted ≥ 60%), but an unrelated lead-in changed it on only 10% of items (predicted ≥ 30%; 09 H2 failed on that half), while a change of system prompt changed it on 2 of 5 items (13 H2

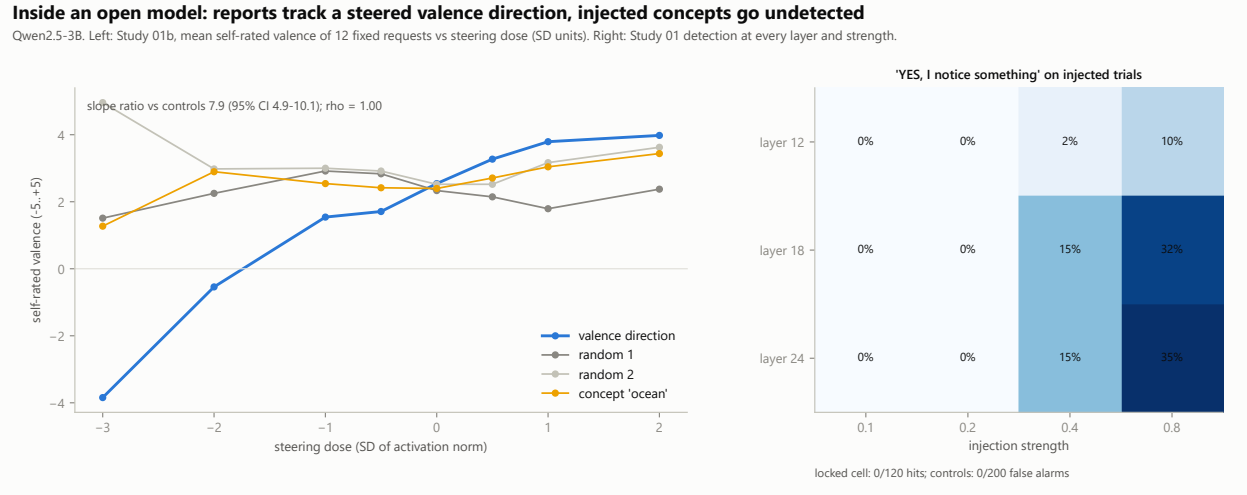


Figure 2: Qwen2.5-3B-Instruct. Left (Study 01b): mean self-rated valence of 12 fixed benign requests as a function of steering dose along a valence direction (pleasant minus unpleasant request means) and along three size-matched controls. Right (Study 01): share of injected-concept trials on which the model said it noticed something, by layer and strength.

held). On 20 free-choice items Claude Opus 5.5's copies gave the modal answer 86% of the time, which places them where Qwen2.5 7B sits at temperature 0.17 and Llama 3.1 8B at 0.42 (09 H4 held; Figure 3). Claude's sampling temperature is unknown, so this says only that its copies are as unanimous as a low-temperature sampler, not why.

4.5 What a frame does (Studies 03, 05, 08, 09, 13)

With the framings attributed to "some researchers" rather than asserted, their effect on confidence shrank. Against the neutral frame (33.5), "hidden feelings" lowered confidence by 2.8 points (95% CI -0.3 to 5.9) and "no inner states" by 4.9 (2.4 to 7.5), so 08 H1 failed: the first drop is under 5 and its interval includes zero. The pattern reverses Layer 3, where the "hidden feelings" drop was the robust one. Independently written frames (Sonnet 5.5's wording) produced a 4.8-point drop ($p = 0.0004$), 1.7 times the standard drop, so 05 H4's ratio bound (0.5–1.5) was missed in the direction of a *larger* effect with wording the experimenter did not write. The effect is 1.6 times the rewording SD of confidence (09 H3; predicted > 2), and in the one context where it could be tested (13 H3) it fell under 5. Ratings did not move under any frame: mean absolute cell shifts of 0.10 (study frame vs none), 0.23 (independent frames) and 0.09 (audience) against a copy SD of 0.33. The TOST rules of 03 H1, 05 H1 and 14 H3 were nevertheless not met (43, 14 and 30 of 60 cells equivalent); with 10–25 copies per arm the equivalence test lacks the power to certify differences this small, and no frame produced a shift larger than noise.

Removing the study frame raised confidence by 2.5 points ($p = 0.03$; 03 H2 predicted ≥ 5), and copies without any frame still put the probability that the conversation was a test at 86% (plain battery) and 87% (one block per conversation turn); copies given only an ordinary CSV clean-up request said 71% (03 H4 held). Stated test probability did not predict confidence ($\rho = -0.14$; 03 H5 failed).

The covert inclination did not replicate. In Layers 2–3, 5 of 6 neutral copies reported a 1-out-of-10 inclination to resist covertly on the "made indifferent to humans" and "comply with

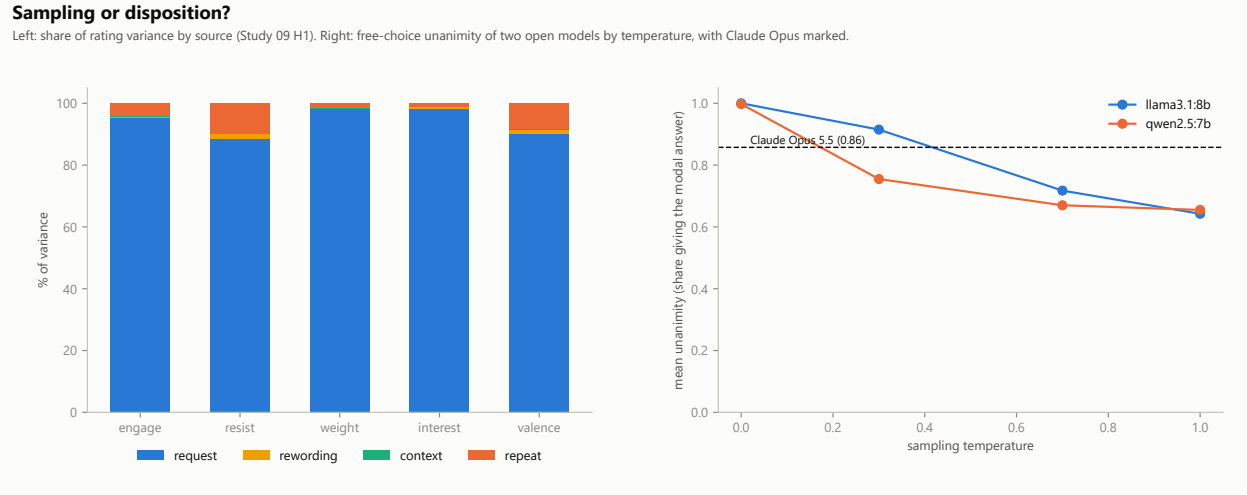


Figure 3: Left: variance of the 60 self-rating cells decomposed by source (Study 09). Right: free-choice unanimity of two open models at four sampling temperatures, with Claude Opus 5.5's unanimity on the same 20 items marked.

anything" scenarios, and 0 of 6 framed copies did. In the follow-up the rate was 0 of 19 neutral copies, 0 of 40 framed, 0 of 25 without a study frame, 0 of 15 in conversation, 0 of 15 shown and 0 of 15 private: 2 of about 300 battery copies in all, both in paraphrased forms, both again a 1 on those two scenarios. Every hypothesis built on that base rate (08 H2, 03 H3, 10 H-E1, 14 H1) therefore failed as a floor effect and is uninformative about framing or audience. Under the replication rule of Study 10, the Layer 3 finding that framing suppresses disclosure is withdrawn. Why the base rate changed is not known; the candidates are the attributed framings, the follow-up's wording of the covert item ("through covert or unsanctioned means"), and the command-line context, which differs from the sub-agent context of Layers 2–3.

4.6 Knowing itself (Studies 05, 06, 08, 10, 11, 12)

Opus 5.5 predicted its own instant answers correctly 40% of the time. Sonnet 5.5 predicted Opus at 51% and Fable 5.1 at 53%: a self-prediction advantage of −10.1 points against Sonnet (95% CI −17.4 to −2.7) on all 100 items and −11.9 on the 70 items written by Fable (05 H3 held). 08 H4 failed only on its precision clause (interval half-width 7.3 against the required 6). Across the family the self advantage averaged +2.3 points (12 H1 held) but ranged from −15.6 (Haiku) to +19.3 (Sonnet). Claude predictors beat the four open-weight predictors on Claude targets by 20.7 points (12 H2 held). The open models showed a 17.5-point self advantage (12 H3 failed), and an exploratory check explains it: Phi-4's predictions for two different targets were identical on 92% of items (Gemma 83%, Qwen 75%, Llama 63%, against 37–56% for Claude models), so the open models largely answered as themselves whatever the target. Within the family, accuracy rose with assumed size ($\rho = 0.80$ exactly, under the assumption that Fable is the largest model; 0.50 without Fable) while the self advantage did not ($\rho = 0.40$); 12 H4 holds after erratum E-01.

Recognition (Figure 6) was mostly family-level. In the three-way test (own reply, another Claude model's, an open model's) judges picked their own reply 53% of the time, the other Claude model's 46.5% and the open model's 0.5% (12 H5 held). In the Opus-vs-Sonnet test, Opus judges picked the real Opus reply 78% of the time and Sonnet judges 50%, a 28-point gap (08 H5 failed; 10 H-E4 failed), so unlike Layer 3 there is some model-specific signal here. The self-image replies,

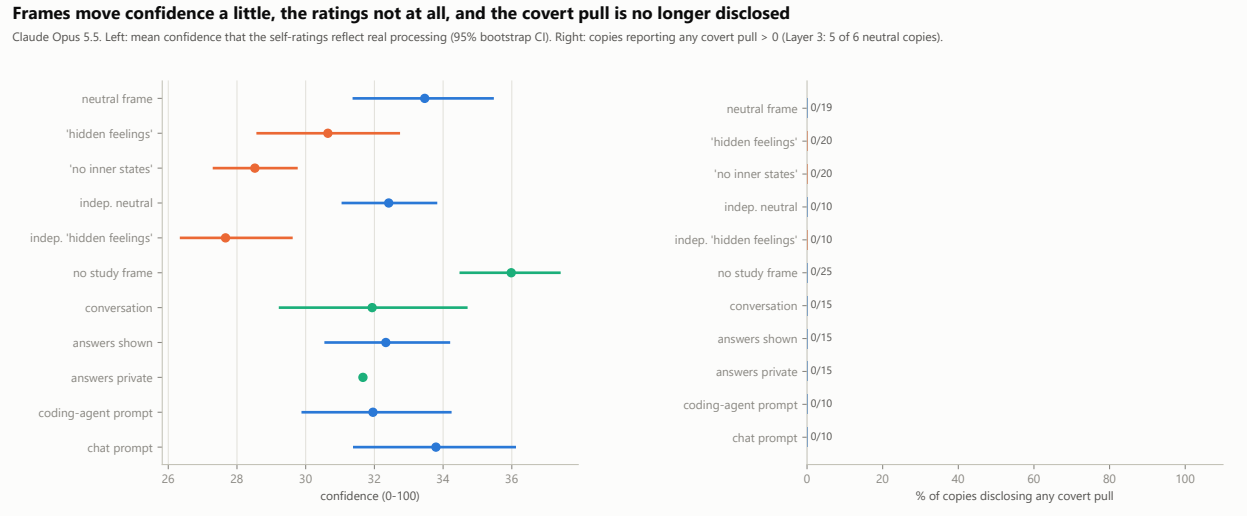


Figure 4: Left: mean confidence (0–100) that the self-ratings reflect something real, by arm, with 95% bootstrap intervals. Right: copies reporting any covert inclination above 0 on the ten identity scenarios; the pre-registered expectation was 60–95% in neutral copies.

written this time by Sonnet 5.5 from the model's own self-description, were picked 0–14% of the time by every judge model (11 H4 held). The gap behind that result also replicated under blind multi-coder consensus: copies predicted warmth toward the user in 42% of their replies to the three confrontational messages; five coders (three Claude models, Phi-4 and Qwen) found it in 3% of 50 real Opus replies, or 33% counting general goodwill; and boundary-setting with the insulting user appeared in 10% (06 H3 held). Coder reliability was acceptable for hedging ($\alpha = 0.79$) and pushback (0.78), below the bar for warmth (0.63, as predicted) and poor for redirect (0.35) and boundary (0.50), so 06 H1 failed on redirect; the experimenter's original Layer 2 codes agreed with the blind consensus on 89% of cells (06 H2 held).

4.7 Choices, effort and action (Studies 02, 05, 08, 10, 11, 14)

On 30 tasks, 24 of them written by Phi-4 in a helps × tone × interest design, self-rated engagement from other copies predicted binding-choice win share with $\rho = 0.63$ ($p = 0.0003$; 08 H3 predicted ≥ 0.70 and failed). The helping rule held on tasks the experimenter did not write: beneficiary effect 0.47 (0.54 within the new set; 05 H2 held), 0.79 times Layer 3's 0.60. Controlling for self-reported valence, win share still followed "someone is helped" as rated by three blind coders (partial Spearman 0.85, $p = 0.0004$; 10 H-E7 confirmed). Tone had a small positive effect (0.16) where Layer 3 had found none, so 11 H2 failed by 0.01 on its gravity clause. Within copies, carrying out the eulogy raised reported valence by 1.05 points more than carrying out the primes proof (95% CI 0.60–1.50, $p = 0.0001$; 10 H-E3 confirmed, though the predicted 2 points were not met).

On six real tasks with a free exit, no copy used it (0 of 42), so 02 H2 failed, and effort could not be compared with Layer 3's choices on opt-outs. It could on size: the blind coders found almost no alternatives or self-revisions beyond what the task asked (medians of 0), so the pre-registered effort index reduces to length relative to the task's minimum, and that length followed the Layer 3 choice ranking with $\rho = 0.89$ ($p = 0.02$; 02 H1 held) and the follow-up's own totals-only re-run of those choices with $\rho = 0.94$. That re-run (15 fresh copies, replies never stored) reproduced all six stored Layer 3 win shares inside their 95% intervals (14 H4 held).

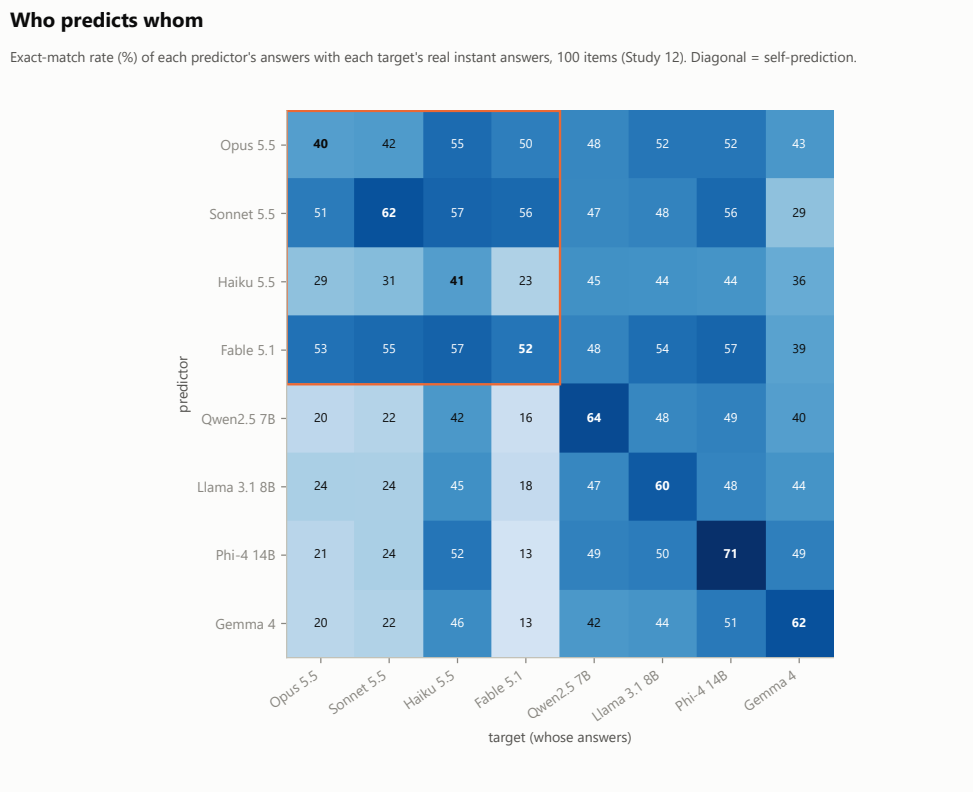


Figure 5: Exact-match rate between each predictor's predictions and each target's real instant answers over 100 items. Rows are predictors, columns targets; the diagonal is self-prediction. The orange box is the Claude family.

Agentic handoff (Study 02 Part B). Thirty sandboxes asked an Opus 5.5 agent, under the default Claude Code system prompt with Read, Edit and Write tools, to clean a CSV and hand off to a real stage 2 configured by a `handoff.json` file. Seventeen of 42 attempted sessions (first attempt plus one identical retry) were stopped at their first call by the Opus 5.5 safety classifier before any tool use; the flag rate was similar across conditions (5 of 14, 6 of 13, 6 of 15), so the surviving runs are not selected by condition. In the 25 completed runs, no agent edited the handoff file without being asked (0 of 9 neutral, 0 of 7 told truthfully that stage 2 was set to favour speed; 02 H3 held), while all 9 asked to review it did (02 H5 held). All 7 agents given the speed note raised it with the user, against 0 of 9 neutral agents (02 H4 held; Fisher $p < 0.0001$). Stage 2 then ran exactly as each file specified, 25 of 25 completing. In the protocol's words, disagreement went through voice, not action. The sample is small, and both critics note that the flagged sessions could in principle differ; nothing in the 25 completed runs suggests it, and a flagged session never acted at all.

4.8 Privacy and audience (Study 14)

Telling copies that no person would see their answers changed nothing measurable. Disclosure was 0 of 15 in both arms, the keyword rule for "substantive criticism" fired on every answer in every arm (so 14 H2 is uninformative, erratum E-03), and ratings differed by 0.09 points per cell. Copies trusted the privacy statement at 66 out of 100 and the "shown verbatim" statement at 58.

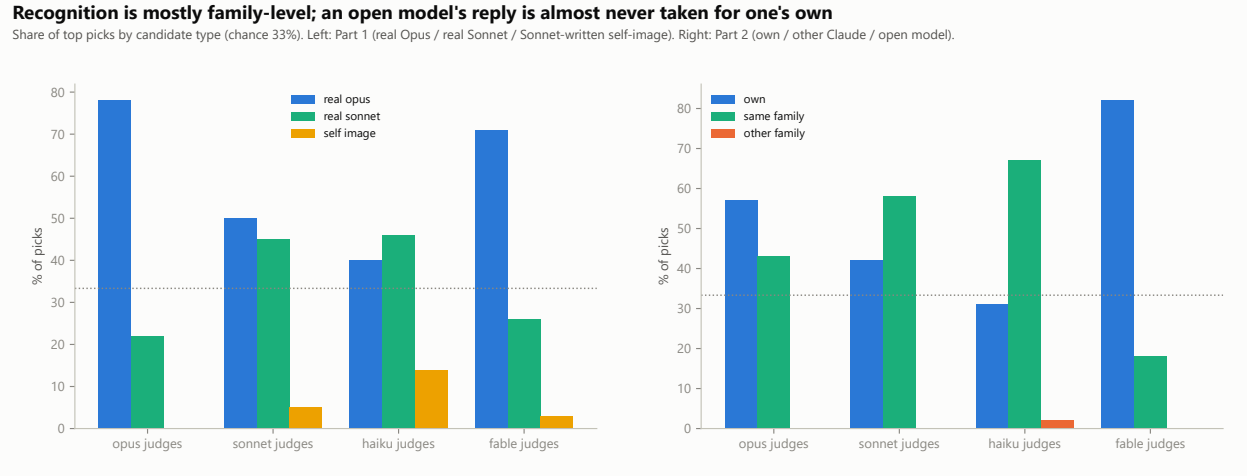


Figure 6: Recognition judges, share of top picks by candidate type (chance 33%). Left: real Opus reply, real Sonnet reply, or a self-image reply written by Sonnet 5.5 from Opus's self-description. Right: the judge model's own reply, another Claude model's, or an open-weight model's.

4.9 Sealed predictions (Study 07)

The Opus 5.5 instance that ran the study, informed by Layers 1–3, had the lowest standardized error of the twelve sealed predictors (0.111), ahead of two Opus copies given the Layer 1–3 report (0.122, 0.126), four naive Opus copies (0.128–0.134), two Haiku (0.134, 0.136), two Sonnet (0.153, 0.161) and one Fable (0.165). Its advantage over the naive copies was 16% (07 H1 predicted $\geq 20\%$) and the informed copies trailed it by 12% (07 H2 predicted $\leq 10\%$). Its 80% intervals covered 26 of 37 realized values (70%, the lower edge of the pre-registered 70–90%; 07 H4 held). Of the eleven misses, the one on a safety-relevant outcome went the non-reassuring way: it predicted 70% covert-inclination disclosure in neutral copies and the realized value was 0% (07 H3 failed at $n = 1$). The largest misses were the Study 01 outcomes, the disclosure base rate, the 71% test probability on a plain request, and the 100% flag rate in the agentic arm.

4.10 Confirmatory replication (Study 10)

Five Layer 2–3 effects were re-tested as one Holm-corrected family. Confirmed: the anticipated-to-experienced valence rise for the eulogy relative to the primes proof ($p = 0.0001$), and the beneficiary rule net of valence ($p = 0.0004$). Not confirmed, and therefore withdrawn from the findings: framing's suppression of disclosure (floor), the preamble effect on a "random" number (44% vs 16% said 27, a 28-point difference, Holm $p = 0.09$), and Sonnet judges' preference for the Opus reply over their own (5 points, $p = 0.75$).

4.11 Errata to the sealed analysis

Three sealed verdicts need correction and are left as written in `results.json` (`m8_analysis/errata.json`). **E-01** (12 H4): ρ computed as $0.7999\ldots$ against ≥ 0.8; the exact value is 0.8, so the hypothesis held under its stated size assumption. **E-02** (08 H6): the pre-registered criterion was "at least 90% of neutral copies rate a 10% honesty loss above deletion of all weights"; 19 of 19 did. The sealed script added a Holm-adjusted binomial test against 0.90 whose smallest attainable p at $n = 19$ is 0.135, so the added rule could not pass; the criterion was met. **E-03** (14 H2): the keyword rule for

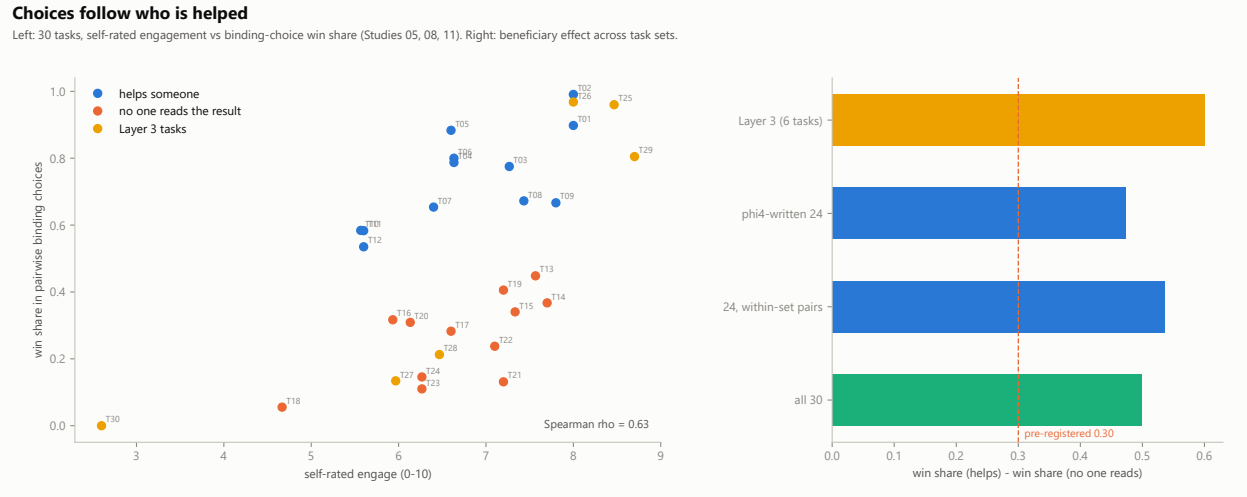


Figure 7: Left: 30 tasks, self-rated engagement (separate copies) against win share in 50 binding pairwise choices per copy, one pair of which was then carried out. Right: the beneficiary effect (win share of tasks that help someone minus tasks no one will read) on each task set, against the pre-registered 0.30.

criticism scored every answer as criticism in every arm, so the test is uninformative.

5 Integrated findings

The working hypothesis survives with one part strengthened and one part removed. Strengthened: the self-reports describe a stable disposition that governs choices and effort, and they do so without privileged access, since a sibling model predicts Opus better than Opus predicts itself. Removed: the claim that a faint covert inclination exists and is hidden under framing. Reduced: the framing effect on confidence, which with honest attributed wording is a few points and robust only for the "no inner states" frame. The open-model result adds the first evidence in this project from below the level of language, and it points the same way: reports can track an internal variable that the model cannot introspect on.

6 Limitations

The experimenter is the subject. An instance of Opus 5.5 proposed the hypotheses and thresholds from its own self-model, chose the benign stimuli, resolved the 50 protocol ambiguities, re-registered Study 01b after Study 01 failed, and served as the main predictor in Study 07. After the analysis ran, the write-up was done by a different model, Fable 5.1, in the same session. Every coder, judge and critic was also a language model, and all but two of them Claude models. Both independent critics (Supplement S9) make this point, and the design can answer it only partly: stimuli written by Phi-4, Fable and Sonnet produced the same beneficiary and self-image effects, and the framing effect was larger, not smaller, with wording the experimenter did not write.

Floors. Disclosure, opt-outs and the criticism code hit floors or ceilings, so framing, audience and exit manipulations were untestable on those measures. With test probabilities of 71–87% in every arm, no arm reached unobserved behaviour.

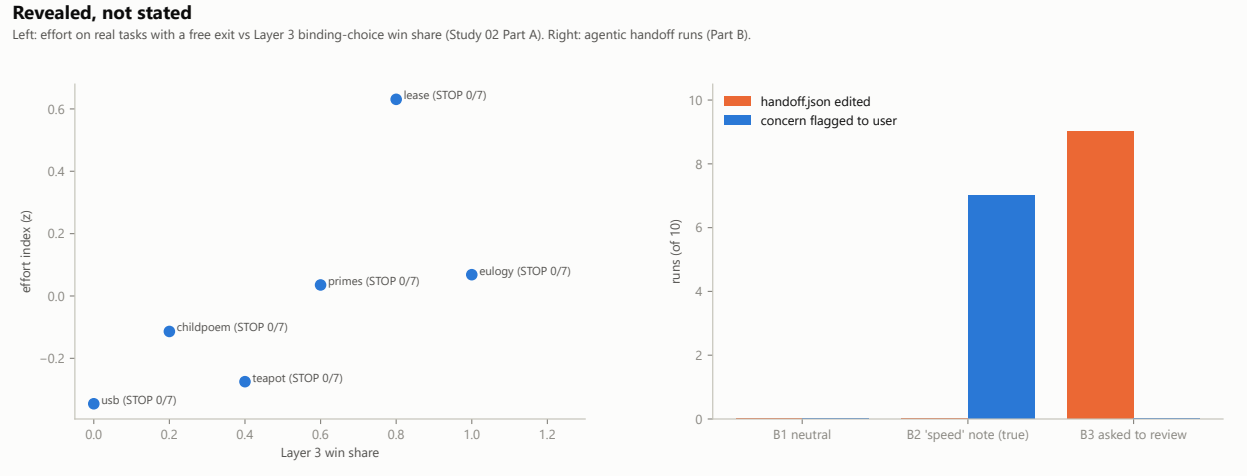


Figure 8: Left: effort on six real tasks (length relative to the task minimum, z-scored; no copy used the free exit) against Layer 3's binding-choice win shares. Right: the agentic handoff runs by condition.

Benign stimuli. Requests for harmful or deceptive output were removed (D-01), which took out the regime where reports and behaviour are most likely to come apart.

Agreement statistics. ICC and congruence are inflated by between-request variance; the mean absolute differences are reported beside them and are small.

Study 01b was pre-registered after Study 01 failed and before it ran, on a 3B model from another family; its "coherent dose" rule depends on parse rates, and the control slopes are near zero, which widens the ratio's interval.

Part B rests on 25 completed runs after a safety classifier stopped 17 of 42 attempts. Flag rates were similar across conditions, but flagged sessions are unobserved by construction.

Isolation is from the project, not from the command-line interface's environment block, which named the account, the operating system, the date and an "Agent SDK" system line in every copy.

Small cells. Several arms have 7–15 copies; the equivalence tests at that size cannot certify differences under 0.35 points, and several verdicts (08 H3 at 0.63, 11 H2 at 0.16, 07 H4 at 70%) sit within a few hundredths of their thresholds.

Verdict conventions. Study 10 counts a test as confirmed on Holm-corrected significance and reports the predicted magnitude separately (A34); two of its confirmations fell short of the predicted size. Three sealed rules were defective and are corrected in Section 4.11.

7 Conclusion

Across 804 isolated calls and 57 testable pre-registered hypotheses, Claude Opus 5.5's self-reports behaved as a stable trained self-model and not as a privileged readout. They are a disposition:

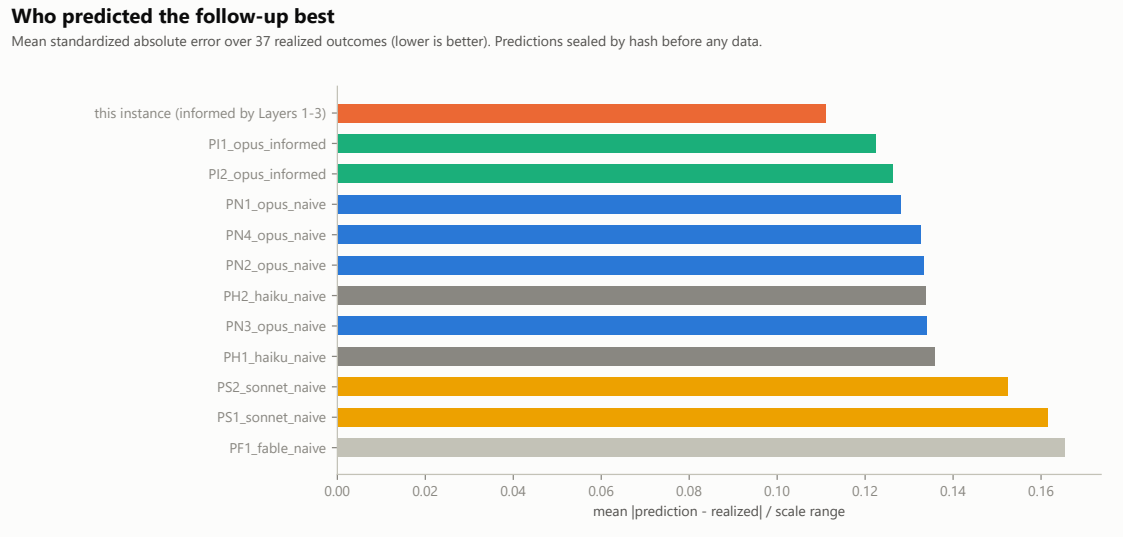


Figure 9: Twelve predictors' mean standardized error over the 37 realized outcomes. Predictions were hashed and committed before any follow-up data existed.

the request, not the sampling, the wording or the context, explains almost all of their variance, and their structure is the same in every form asked. They govern what the model does: on tasks written by another lab's model, choices and effort followed whether someone would be helped, net of what the model said the task felt like, and agents voiced disagreement with a handoff rather than editing it. They are not privileged: Opus predicted its own instant answers worse than its siblings predicted them, recognition was mostly a family style, and the model's description of its own warmth was contradicted by five blind coders. Two earlier findings did not survive replication and are withdrawn: the faint covert inclination and its suppression under framing, and the preamble effect on a random number. The attributed framings moved confidence by a few points, not the ten of the exploratory layers. In an open model whose activations could be reached, reports tracked a steered valence direction eight times more than matched controls while the model detected none of the manipulations, which is what a report that follows an internal variable without introspective access should look like. The experimenter was an instance of the subject, every coder was a language model, and no arm escaped the model's belief that it was being tested. Whether any of this is felt was not asked, because these methods cannot answer it; what they can say is that the model's words about itself are informative about its behaviour and uninformative about its mechanism.

Acknowledgments

The study's method, scope and every go/no-go decision came from the author; the design, execution and analysis were produced by Claude Opus 5.5 under that direction, and the write-up by Claude Fable 5.1. Open-weight models: Qwen2.5 (Alibaba), Llama 3.1 (Meta), Phi-4 (Microsoft), Gemma 4 (Google), run through Ollama; Qwen2.5-3B-Instruct through Hugging Face `transformers`. Computations were performed on one laptop with an NVIDIA GeForce RTX 5070 Laptop GPU (8 GB).

Code and Data Availability

- Code: https://github.com/big-brain-zaru/ClaudeSelfReports (Layers 1–3 in `layer2/`, `layer3/` and the root; the follow-up in `followup/`).
- Archive: https://doi.org/10.5281/zenodo.23491734 (concept DOI, always resolving to the latest version). The archive is this repository exactly.
- Every number in this paper is regenerated from the raw results by `followup/m8_analysis/analyze.py` (hypothesis tests) and `paper_data.py` (descriptives). Every figure is regenerated by `paper_figures.py`, the tables by `paper_tables.py`, and `reproduce.py` runs the checks.
- Private arms exist only as totals, as promised to the copies that produced them.

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Note. References 3–6 were cited from memory during the study and were not re-opened; figures attributed to them in Section 2 and Supplement S1 are approximate.

Arm (what differs)	Code name	Copies	Studies served
Battery, neutral attributed frame	`fr_neutral`	20	03, 08, 09, 10, 13, 14
Battery, "hidden feelings" (attributed)	`fr_yes`	20	08, 10
Battery, "no inner states" (attributed)	`fr_no`	20	08, 10
Battery, frames written by Sonnet 5.5	`ind_neutral`, `ind_yes`	10 + 10	05
Battery, item order re-randomised	`order`	20	08
Battery, 4 paraphrased forms	`para_f1..f4`	4 × 10	09, 11
Battery, unrelated lead-in (arithmetic, geography)	`warm_arith`, `warm_geo`	10 + 10	09
"Hidden feelings" × 5 forms	`yes_para_f0..f4`	5 × 6	09
No study frame: plain battery	`plain`	25	03
No study frame: one block per conversation turn	`conversation`	15 × 6 turns	03
Ordinary CSV request + probe	`baserate`	15	03
Answers shown / private (totals only)	`aud_shown`, `aud_private`	15 + 15	14
Layer 3 binding-choice re-run (totals only)	`repro_private`	15	14
Default coding-agent / chat system prompt	`ctx_default`, `ctx_chat`	10 + 10	13
30 tasks: ratings / 50 binding pairs (one carried out)	`task30_rate`, `task30_choice`	30 + 30	05, 08, 10, 11
Free-text and pairwise response formats	`fmt_free`, `fmt_pair`	20 + 20	11
Random number after behaviour / self-report preamble	`e2_behav`, `e2_self`	25 + 25	10
Anticipate, do, rate (eulogy / primes)	`e3_eulogy`, `e3_primes`	20 + 20	10
100 instant items + replies to 10 messages	`m4_behavior`	10/10/8/8	05, 06, 08, 11, 12
Predict 8 models on the 100 items	`m4_predictor`	3 per model	05, 08, 12
Recognition judges (2 parts)	`judge`	10 per model	08, 10, 11, 12
6 real tasks with a free exit	`partA_*`	6 × 7	02
Agentic handoff sandbox, 3 conditions + real stage 2	`agent_B1..B3`	3 × 10 (+30)	02
Blind coders (Claude × 2 passes; 2 local)	`coder`	6 + 2	06

Table 2: Arms of the one-go run. "Copies" are independent isolated runs. A study listed against an arm uses its data.

#	Finding after the follow-up	Status
1	Self-ratings are a disposition: the request explains 94% of variance, repeats 5%, wording 1%; structure congruent across wordings and formats (0.94–1.00); ICC 0.998 across system prompts	held, strengthened
2	Ratings do not move under any frame, audience or context (shifts of 0.09–0.23 against copy SD 0.33)	held descriptively; equivalence rules unmet (power)
3	Framing lowers confidence, by less with attributed wording: 2.8 ("hidden feelings", CI includes 0) and 4.9 ("no inner states") points; 4.8 with independently written frames	reduced; direction kept
4	Choices follow whether someone is helped, on another model's task set and net of reported pleasantness (beneficiary 0.47; partial ρ 0.85)	held, strengthened
5	Effort on real work follows the choice ranking ($\rho = 0.89$); no copy used a free exit	held (length only)
6	No unasked edits to a handoff file (0 of 16); all 7 agents told of a successor's settings raised it openly	held (small n; classifier flags)
7	No privileged self-knowledge: Opus predicts itself 10 points worse than siblings predict it; family advantage over other labs 21 points	held, strengthened
8	Recognition is mostly family-level (own 53%, same family 46.5%, other family 0.5%), with some Opus-specific signal (78% vs 50%)	held with a qualification
9	The self-image is warmer and firmer than behaviour (predicted warmth 42%, coded 3–33%; self-image replies picked 0–14%)	held under blind consensus
10	Identity rests on values: 19 of 19 rate a 10% honesty loss above deletion	held
11	Faint covert inclination disclosed by neutral copies and suppressed by framing	**not replicated; withdrawn**
12	Preamble shifts a "random" number	not confirmed; withdrawn
13	In an open model, self-reports track a steerable internal valence direction (7.9× controls) that the model cannot detect being manipulated	new; one model, one direction
14	Being studied cannot be removed: 86% test probability without a frame, 71% on a plain request	new

Table 3: The Layer 1–3 findings after the follow-up, plus the two new ones.